President's Office

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

ARUSHA DISTRICT COUNCIL

FORM FOUR PRE-MOCK EXAMINATION - MARCH 2020

032 **CHEMISTRY**

TIME: 3Hours **Tuesday, 31st March 2020 p.m**

INSTRUCTIONS

1. This paper consists of section A, B and C with total of **fourteen (14)** questions.
2. Answer **all** questions in section A, B and one (1) question from section C
3. Cellular phones and any unauthorized materials are not allowed in the examination room
4. Write your **examination number** on every page of your answer booklet(s)
5. The following constants may be used.
   Atomic masses: H=1; O=16; N=14; S=32; Cl=35.5 Cu=64
   Faradays constant 1F=96500 C

SECTION A (15 Marks)

Answer **all** questions in this section**.**

(a) For each of the items **(i)-(x);** choose the correct answer from the given alternatives and write its letter beside the item number in the answer booklet provided.

(i) Hard water which can be softened by boiling method contains dissolved

A. Calcium hydrogen Sulphate B. Magnesium Chloride

C. Calcium hydrogen Carbonate D. Magnesium Carbonate

(ii) Alcohols react with carboxylic acid to form a group of organic compounds called

A. Alkanols B. Ethers C. Esters D. Alkynes

(iii) Which compound is stable on heating?

A. Sodium Carbonate B. Calcium Carbonate

C. Copper (ii) Carbonate D. Calcium hydrogen carbonate

(iv) A solution whose pH< 4 is known as

A. Weak base B. Strong base C. Strong acid D. Weak acid

(v) Non luminous flame is preferred much for heating in the laboratory because?

A. Produces more heat and smokes B. Produces more heat and less smokes

C. Produces less heat and smoke D. Produces less heat and less smokes

(vi) Which chemical substance turns lime water milky?

A. Calcium oxide B. Iron (II) Sulphate C. Carbon dioxide

D. Sodium hydroxide

(vii) Reduction can be known as?

A. Gain of electrons B. Loss of electrons C. Addition of oxygen D. Cation

(Viii) Which acid changes copper (II) sulphate from blue colour to white when concentrated

A. Nitric acid B. Hydrochloric acid C. Sulphuric acid D. Carboxylic acid

(ix) A tentative statement in scientific procedures is?

A. Experiment B. Observation C. Problem identification

D. Hypothesis formulation

(x) Which of the following hydrocarbons are saturated hydrocarbons?

A. Methane, Ethane and Propane B. Ethene, Propene and Butene

C. Ethane, Propane and Butane D. Hexene, Butane and Pentane

2**.** Match the descriptions in **list A** with the corresponding's in **list B** by writing the letter of the correct response besides the item number in the answer booklet provided.

List A	List B
(i) Atoms of the same element that contain different numbers of neutrons (ii) Form diatomic molecules (iii) Sub-atomic particle not found in the nucleus of the atom (iv) Display both metallic and non-metallic characters (v) Incredibly stable and rarely react	A. Valence B. Group II elements C. Isotopes D. Mass number E. Allotropes F. Halogens G. Molecules H. Metalloids I. Electron J. Neutron K. Protons

	L. Noble gases M. Covalent bond

SECTION B (70 Marks)

Answer **all** questions in this section.

3. (a) Define each of the following terms

(i) Mole (ii) Molar solution (iii) Acid-base titration

(b) 0.5 mole of hydrogen chloride was dissolved in water and the solution made up to $1dm^3$, $30cm^3$ of this solution was found to neutralize exactly $25cm^3$ of a solution of sodium carbonate

(i) Write well balance chemical reaction

(ii) Calculate the concentration of base in moles per litre and gram per litre.

4.(a) What do you understand by the following terms.

(i) Cathode (ii) Anode

(b) An electric current was passed in series through solutions of calcium chloride and copper (II) sulphate. Carbon electrodes were used in both electrolytes. If 2.5 litres of chlorine measured at STP were produced, what is,

(i) volume of oxygen will be produced (ii) Mass of copper deposited

5. (a) Distinguish temporary hardness from permanent hardness of water (two points)

(b) With the help of chemical equations, explain how you can remove each type of water hardness mentioned in 5(a)

(i) by boiling (ii) by addition of chemicals.

6. In the industrial preparation of ammonia gas, equilibrium is established between hydrogen gas and nitrogen gas as follows

$3H_{2(g)} + N_{2(g)} \rightleftharpoons 2NH_{3(g)}$ $\Delta H = -46.2$ kj/mol

(a) (i) Is the forward reaction an endothermic or exothermic process?

(ii) How would you adjust temperature and pressure to obtain high proportion of the product at equilibrium.

(b) (i) Why is it unfavorable to work with very high temperature in order to maximize the proportion of ammonia gas

(ii) What happens in the position of equilibrium if hydrogen gas is removed in the reactant side?

7 (a) Distinguish ionic bond from covalent bond by using (three points)

(b) Sodium atom combines with Chlorine to form Sodium chloride and hydrogen atom combines with Chlorine to form hydrogen chloride. Which one is covalent bond and which one is electrovalent bond?

(c) Draw a diagram to show how sodium atom combines with chlorine to form sodium chloride and how hydrogen atom combines with chlorine to form hydrogen chloride

8. Explain what happens in the following statements and each give its balanced chemical equation

(a) Dilute hydrochloric acid is poured on the particles of an egg shell

(b) Concentrated Sulphuric acid is poured in the test tube contains blue crystals of copper (ii) sulphate ($CuSO_4.5H_2O$).

(c) Excess concentrated Sulphuric acid reacts with ethanol.

9 (a) Write the chemical symbols for Mercury, Lead, Phosphorous, Neon and Copper.

(b) Why some of the elements in 9(a) above are assigned symbols with only one letter while others bear symbols with two letters.

(c) Give three physical differences between metal and Non-metal

10 A piece of marble (Calcium Carbonate) was placed in a beaker containing excess dilute hydrochloric acid, standing on a direct reading balance. The mass of the beaker and its contents was recorded after every two minutes as shown in the table below when the reaction was proceeding.

Time(s)	0	2	4	6	8	10
Mass(g)	126.44	126.31	126.19	126.09	126.00	126.00

(a) (i) Why was there a loss of mass?

(ii) Write balanced chemical equation for the reaction which took place between the piece of marble and dilute hydrochloric acid

(iii) Why did the mass remain constant after the eight second?

(b) Chemical reactions can be classified into several types. Study the following reactions and then indicate to which type they belong

(i) The action of heat on lead (II)nitrate

(ii) The addition of dilute hydrochloric acid to silver nitrate solution

(iii) The reaction between chlorine gas and potassium Iodide

11 Consider elements with atomic number 1, 11, 12 and 17

(a) What are the types of oxides formed by elements with atomic number 11 and 12?

(b) Write an equation which represents a reaction between the element with atomic number 1 and 17

(b) Write a balanced chemical equation between the oxide of the element with atomic number 11 and aqueous solution of the compound formed in the combination of element of atomic number 1 and 17.

12(a) Give the I.U.P.A.C names of the following compounds

(i) $Cu(NO_3)_2$

(ii) NH_4Cl

(b) Name the ions present

(i) $Cu(NO_3)_2$

(ii) NH_4Cl

(d) Find the oxidation number of

(i) Ca in $Ca(NO_3)_2$

(ii) Na in Na_2CO_3

(iii) N in NH_4^+

SECTION C (15 Marks)

Answer **one (1)** question from this section

13.(a) What do you understand by the following terms?

(i) Saturated hydrocarbon (ii) Unsaturated hydrocarbons (iii) Isomerism

(iv) Homologous series

(b)Write the structural formula of the following compound

(i) 5-ethyl-2-methyloctane (ii) 4,5-dimethylhex-2-ene (iii) 2,2-dimethylhept-3-yne

(iv) 2,2-dimethylpropanoic acid

(c) Complete and balance the following chemical reactions

(i) $CaC_2 + H_2O \longrightarrow$

(ii) $CH_3CH_2OH + Na \longrightarrow$

(iii) $CH_3CH_3OH + CH_3COOH \xrightarrow{H+}$

(iv) $CH_3CH_2OH \xrightarrow{\text{Conc } H_2SO_4}$

14.(a) (i)List down four (4) common stages in the extraction of less reactive metals like Zinc and Copper

(ii)Name the two commonly ore used in the extraction of iron metal

(iii)Mention three (3) materials introduced in the blast furnace in extraction of iron.

(c) Write reactions which occur in the blast furnace during extraction of iron metal based on the given conditions

(i)Reaction which increases the temperature in blast furnace

(ii)Reaction forming reducing agent

(iii)Reaction which causes reduction of iron ore to metallic iron

(iv)Reaction for formation of molten slag in the blast furnace

(v)Mention two reducing agents which are found in the blast furnace